

Chapter 6: Magnetic Mineralogy

Essentials of Paleomagnetism

Based on Tauxe et al.

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Why Magnetic Mineralogy?

- Every paleomagnetic study must identify **what carries the remanence** and **how the rock was magnetized**
- Iron is the most abundant transition element in the solar system
- Key magnetic mineral groups:
 - **Iron-oxides**: magnetite, maghemite, hematite
 - **Iron-oxyhydroxides**: goethite, ferrihydrite
 - **Iron-sulfides**: greigite, pyrrhotite
- Titanium substitution profoundly alters magnetic properties

Solid Solutions in Iron-Oxides

- **Solid solution:** one or more atom types partly substituted without changing crystal structure
- Ti^{4+} commonly substitutes for Fe in iron oxides
- Ti^{4+} has **no unpaired spins** → changes magnetic properties
- Two critical solid solution series:
 - ① **Ulvöspinel–Magnetite** (titanomagnetites)
 - ② **Ilmenite–Hematite** (titanohematites)
- Both crystallize at $\sim 1300^\circ\text{C}$
- Complete solid solution above $\sim 600^\circ\text{C}$ (titanomagnetites) and $\sim 800^\circ\text{C}$ (titanohematites)

Phase Diagrams for FeTi Oxides

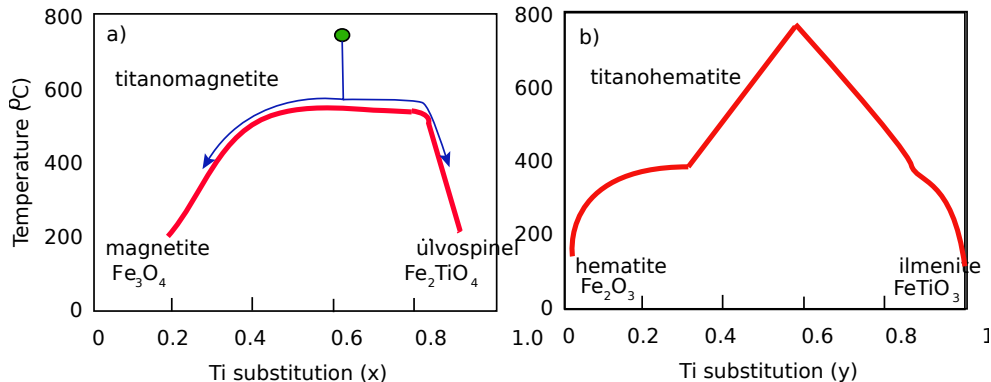


Figure: Phase diagrams showing exsolution with cooling. a) Titanomagnetite series. b) Titanohematite series.

Exsolution

- As temperature drops below the solvus, intermediate compositions become unstable
- Cation diffusion produces Ti-rich and Ti-poor **lamellae**
- Rapid cooling → metastable compositions preserved
- Paleomagnetic importance:
 - Different compositions = different magnetic properties
 - Lamellae reduce effective grain size → enhanced stability

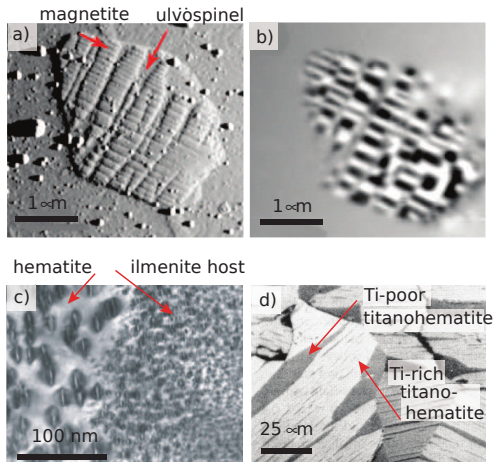


Figure: a) AFM of magnetite in pyroxene. b) MFM of magnetite. c) TEM of hematite. d) TEM of hematite.

Ternary Diagram for Iron-Oxides

- Apices: Fe^{2+} , Fe^{3+} , Ti^{4+}
- Solid arrows: increasing Ti substitution
 - x : Ti in titanomagnetites
 - y : Ti in titanohematites
- Dashed arrows: direction of increasing oxidation (z)
- x, y range from 0 to 1

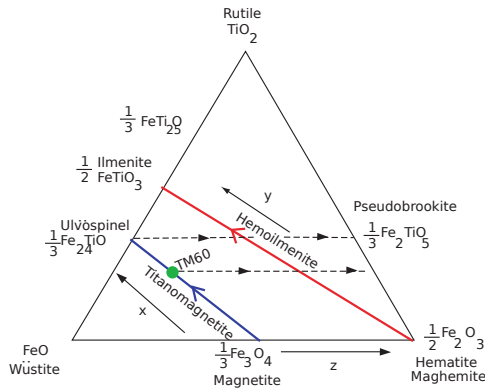


Figure: Fe^{2+} - Fe^{3+} - Ti^{4+} ternary diagram showing titanomagnetite and titanohematite series.

Titanomagnetite Structure

- Magnetite (Fe_3O_4): inverse spinel (AB_2O_4)
- O atoms form FCC lattice; cations in tetrahedral (A) and octahedral (B) sites
- Cation distribution: $\text{Fe}^{3+}|\text{Fe}^{3+}\text{Fe}^{2+}|\text{O}_4$
- A and B sublattices coupled **antiparallel** $\rightarrow$ ferrimagnetic
- Net moment: $(9 - 5) = 4 m_b$ per molecule
- Ti^{4+} substitution:

	Magnetite ($x = 0$)	Ulvöspinel ($x = 1$)
Structure	$\text{Fe}^{3+} \text{Fe}^{3+}\text{Fe}^{2+} \text{O}_4$	$\text{Fe}^{2+} \text{Fe}^{2+}\text{Ti}^{4+} \text{O}_4$
Behavior	Ferrimagnetic	Antiferromagnetic

Effect of Ti Substitution on Titanomagnetites

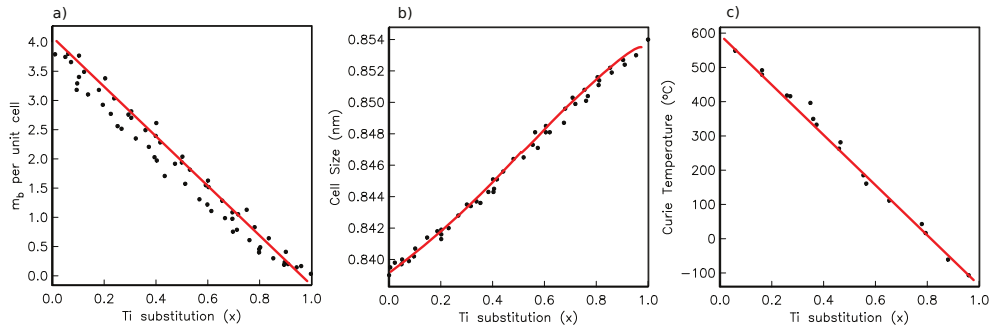


Figure: Variation with Ti substitution (x): a) Saturation magnetization decreases. b) Cell size increases. c) Curie temperature decreases.

Titanomagnetite: Key Points

- **Occurrences:** primary in igneous rocks; detrital in sediments; biogenic; authigenic
- Large $M_s \rightarrow$ shape anisotropy dominates for even slight non-equancy ($\sim 10\%$)
- **Verwey transition** (~ 120 K): abrupt loss of magnetization
 - Diagnostic for magnetocrystalline-anisotropy-dominated remanence
 - Sensitive to oxidation; suppressed by maghemitization
- Natural impurities (Al^{3+} , Mg^{2+} , Cr) also affect properties
 - $0.1 \text{ Al}^{3+} \rightarrow 25\%$ reduction in M_s , $\sim 50^\circ\text{C}$ drop in T_c

Hematite ($\alpha\text{Fe}_2\text{O}_3$)

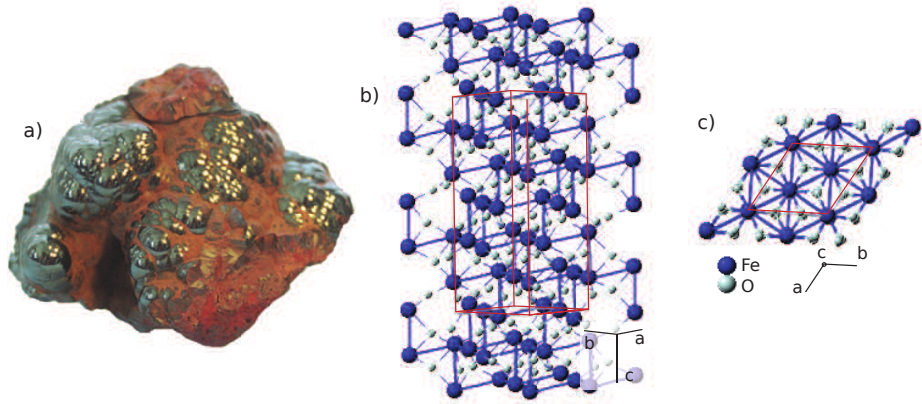


Figure: a) Kidney ore hematite. b–c) Crystal structure (corundum-type); *c*-axis perpendicular to basal plane.

Hematite Properties

- Corundum structure, rhombohedral, pseudocleavage $\perp$ c -axis
- **Antiferromagnetic** with weak parasitic ferromagnetism
 - Spin-canting or defect ferromagnetism
- Néel temperature $\approx 685^\circ\text{C}$
- **Morin transition** ($\sim -10^\circ\text{C}$):
 - Above: magnetization in basal plane
 - Below: spin-canting disappears; magnetization $\parallel$ c -axis
 - Most low-T hematites dominated by defect moments $\rightarrow$ no Morin transition observed
- Dominates magnetic properties of **red beds**
- Very high coercivity; very low M_s ($\sim 0.4 \text{ Am}^2\text{kg}^{-1}$)

Ti Substitution in Titanohematites

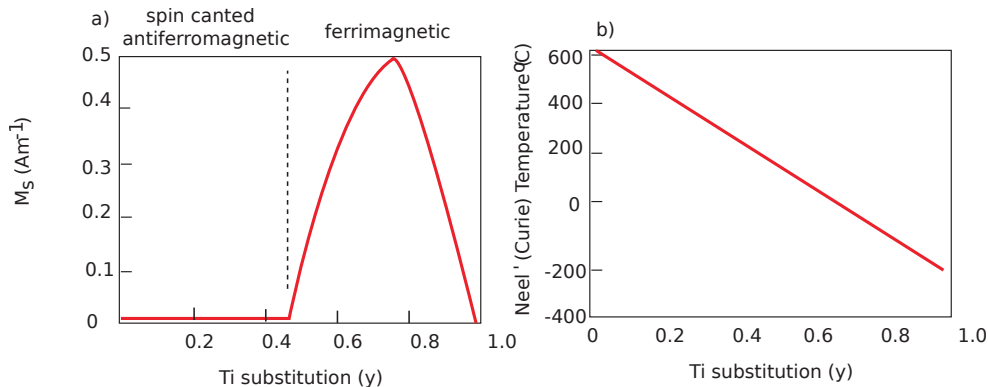


Figure: a) Saturation magnetization vs. y : ferrimagnetic above $y \approx 0.45$. b) Néel temperature decreases with Ti substitution.

Self-Reversal in Titanohematites

- For intermediate y , exsolution produces Ti-rich and Ti-poor lamellae
- Ti-poor bands: lower M_s , higher $T_N \rightarrow$ magnetize first
- Ti-rich bands: higher M_s , lower $T_N \rightarrow$ magnetize later in the demagnetizing field of Ti-poor bands
- Result: Ti-rich bands acquire remanence **antiparallel** to applied field
- Because Ti-rich bands have higher M_s , net NRM is **self-reversed**
- Fortunately very rare in nature

Low-Temperature Oxidation to (Titano)maghemite

- Single-phase spinel $\rightarrow$ cation-deficient spinel
- Fe^{2+} diffuses to surface, converts to Fe^{3+}
- Consequences of increasing z :
 - M_s decreases
 - Cell size shrinks
 - T_c increases
 - Characteristic surface cracking
- Maghemite ($\gamma\text{Fe}_2\text{O}_3$) is metastable
- Inverts to hematite ($\alpha\text{Fe}_2\text{O}_3$) by $\sim 350^\circ\text{C}$

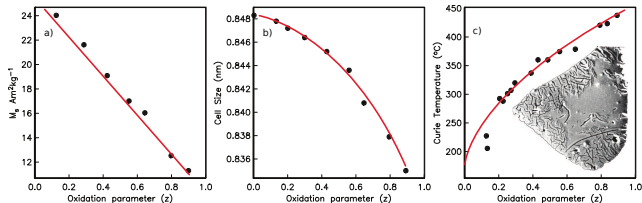


Figure: Oxidation (z) effects on TM60: a) M_s , b) cell size, c) T_c . Inset: cracking from volume change.

Verwey Transition Suppression by Maghemitization

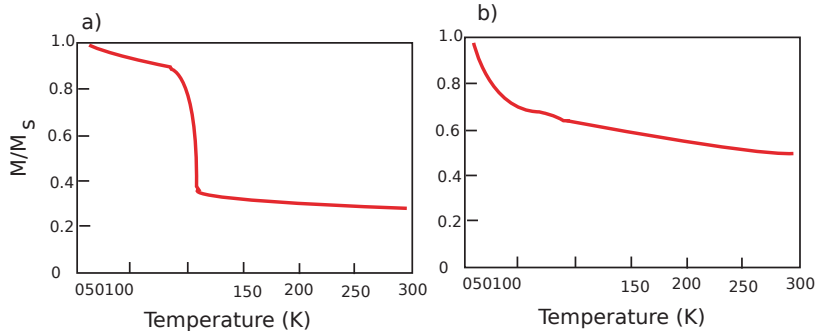


Figure: a) Saturation remanence warming curve for stoichiometric 37 nm magnetite showing the Verwey transition. b) Partially oxidized magnetite: transition suppressed.

- High-temperature oxidation during initial cooling of lava
- Titanomagnetite + $O_2 \rightarrow$ low-Ti magnetite + ilmenite intergrowths
- Reaction: $Fe_2TiO_4 + O_2 \rightleftharpoons Fe_3O_4$
- Explains why ulvöspinel–magnetite intergrowths are rare despite solid solution
- Distinguishes from **maghemitization** (low-T, single-phase oxidation)

Goethite, Greigite, and Pyrrhotite

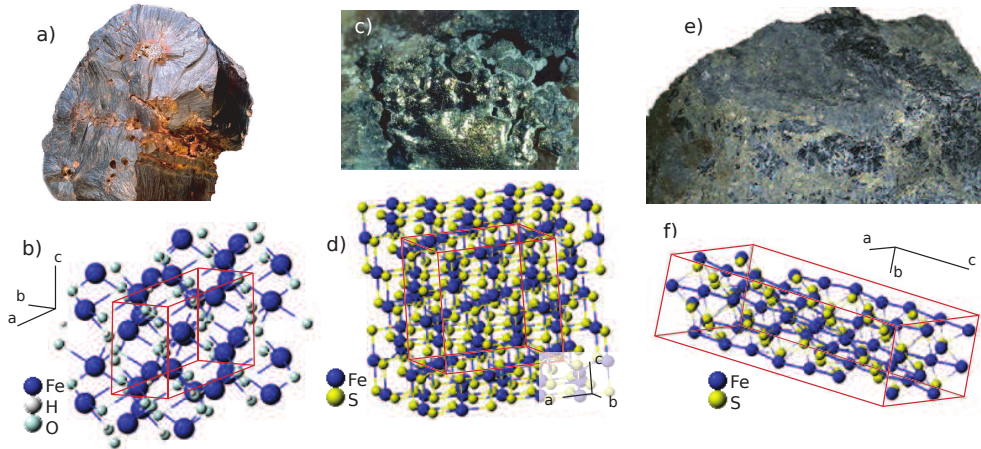


Figure: a–b) Goethite photo and crystal structure. c–d) Greigite photo and structure. e–f) Pyrrhotite photo and structure.

Goethite (αFeOOH)

- Most common magnetic iron-oxyhydroxide
- Antiferromagnetic with **defect magnetization**
- Forms as weathering product or direct precipitate
- Metastable: dehydrates to hematite by $\sim 325^\circ\text{C}$
- Very high coercivity but low T_N ($\sim 100\text{--}150^\circ\text{C}$)

Iron-sulfides (greigite Fe_3S_4 , pyrrhotite Fe_7S_8):

- Ferrimagnetic; occur in **reducing environments**
- Both tend to oxidize to iron oxides + paramagnetic pyrite
- Greigite: max unblocking $T \approx 330^\circ\text{C}$

Pyrrhotite Thermomagnetic Behavior

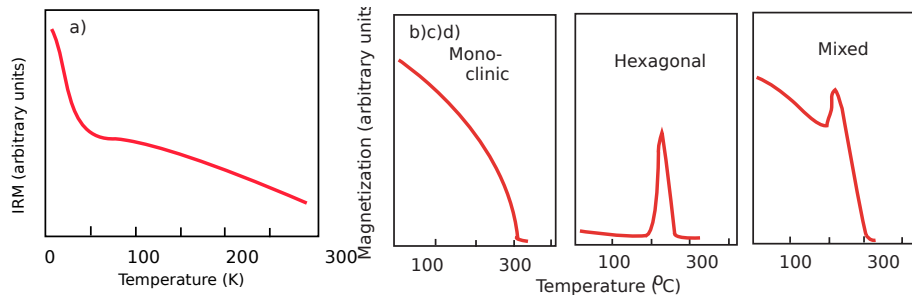


Figure: a) Low- T transition in monoclinic pyrrhotite (~ 35 K). Thermomagnetic curves for b) monoclinic ($T_c \approx 325^\circ\text{C}$), c) hexagonal (peak near 200°C), d) mixture.

FeTi Oxides in Igneous Rocks

- Composition depends on bulk chemistry, fO_2 , cooling rate
- More abundant in mafic (basalts) than silicic (rhyolites) lavas
- Ti content generally lower in more silicic melts
- Tholeiitic basalts: $0.5 < x < 0.8$; typically TM60
- Rhombohedral phases: $0.05 < y < 0.3$
- Slowly cooled rocks: exsolution or deuteritic oxidation
- Rapidly quenched: metastable compositions preserved

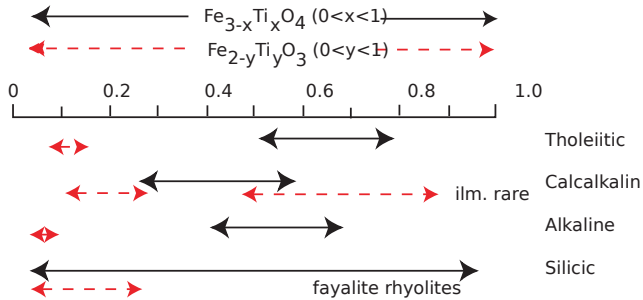


Figure: FeTi oxide compositions in igneous rocks. Black lines: titanomagnetite; red dashed: hemoilmenite.

Sedimentary Magnetic Mineralogy

- Igneous rocks are the ultimate source, but biological and diagenetic processes modify the minerals
- Detrital (titano)magnetite may alter under changed pH/redox conditions
- With depth: organic carbon breakdown → reducing conditions
 - Magnetite dissolution
 - Authigenic sulfide production
 - Decreasing susceptibility, IRM, ARM with depth
- A virtual “rainbow” of magnetic mineralogies in sediments

Biogenic Magnetite

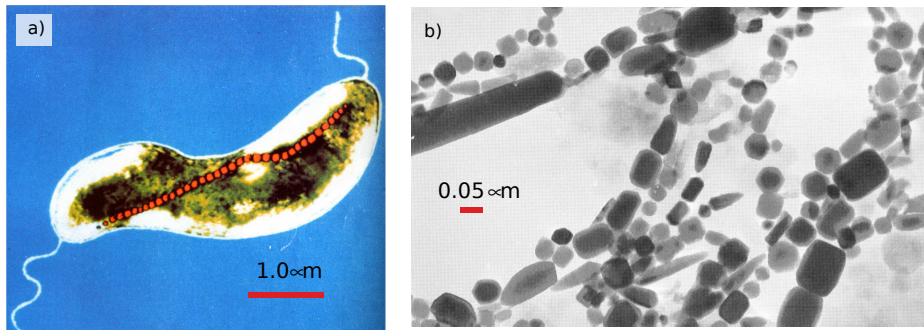


Figure: a) Intact magnetosome in a living magnetotactic bacterium. b) Magnetosome chains recovered from ODP Site 1006D, Bahamas.

- Magnetotactic bacteria produce single-domain magnetite
- Sizes and shapes fall in the SD range on the Evans diagram
- Common in sediments; role in NRM still poorly understood

Diagnostic Tools: Thermomagnetic Curves & Microprobe

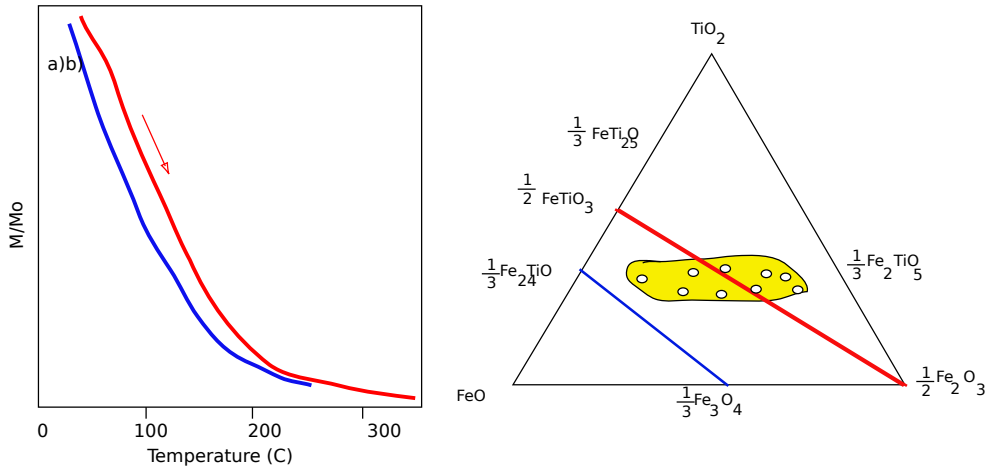


Figure: a) Thermomagnetic run: M/M_0 vs. T (red = heating, blue = cooling). b) Electron microprobe data on TiO_2 -FeO- Fe_2O_3 ternary diagram.

Curie Temperature as a Diagnostic

- Curie (or Néel) temperature is a primary identifier
- Key values:
 - Magnetite: 580°C
 - Hematite (T_N): 675°C
 - TM60: 150°C
 - Goethite (T_N): 100–150°C
 - Pyrrhotite: ~325°C
 - Greigite: ~330°C
- Coercivity spectra provide additional constraints

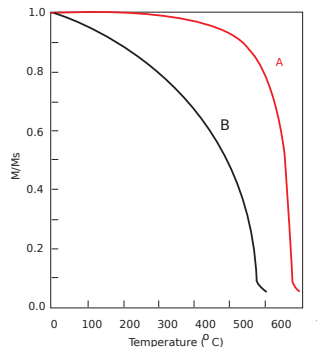


Figure: Curie temperature curves for two samples A and B.

Key Mineral Properties

Mineral	T_c or T_N	M_s ($\text{Am}^2\text{kg}^{-1}$)	Type
Magnetite	580°C	92	Ferrimagnetic
Maghemite	590–675°C	74	Ferrimagnetic
TM60	150°C	24	Ferrimagnetic
Hematite	675°C	0.4	Canted AFM
Goethite	100–150°C	<1	Defect AFM
Pyrrhotite	~325°C	0.4–20	Ferrimagnetic
Greigite	~330°C	~25	Ferrimagnetic

Key diagnostic transitions:

- Verwey transition (~ 120 K) $\rightarrow$ magnetite (suppressed by oxidation)
- Morin transition (~ 263 K) $\rightarrow$ hematite
- 35 K transition $\rightarrow$ monoclinic pyrrhotite

Chapter Summary

- 1 **Titanomagnetites** ($\text{Fe}_{3-x}\text{Ti}_x\text{O}_4$): Ti substitution decreases M_s and T_c , increases cell size
- 2 **Titanohematites** ($\text{Fe}_{2-y}\text{Ti}_y\text{O}_3$): transition from canted AFM to ferrimagnetic at $y \approx 0.45$; can self-reverse
- 3 **Oxidation**: converts magnetite $\rightarrow$ maghemite (increases T_c , decreases M_s , suppresses Verwey transition)
- 4 **Deuteric oxidation**: high- T process producing magnetite + ilmenite from titanomagnetite
- 5 **Goethite**: low T_N , high coercivity; dehydrates to hematite
- 6 **Iron-sulfides**: ferrimagnetic in reducing environments; diagnostic thermomagnetic signatures
- 7 **Sediments**: complex mineralogy shaped by biology, diagenesis, and redox conditions